

§3 利用柱坐标和球坐标计算三重积分

- 👉 利用柱坐标计算三重积分
- 👉 利用球坐标计算三重积分

一、利用柱坐标计算三重积分

1. 柱面坐标

设 $M(x, y, z) \in \mathbb{R}^3$, 将 x, y 用极坐标 ρ, θ 代替, 则 (ρ, θ, z) 就称为点 M 的柱坐标. 直角坐标与柱面坐标的关系:

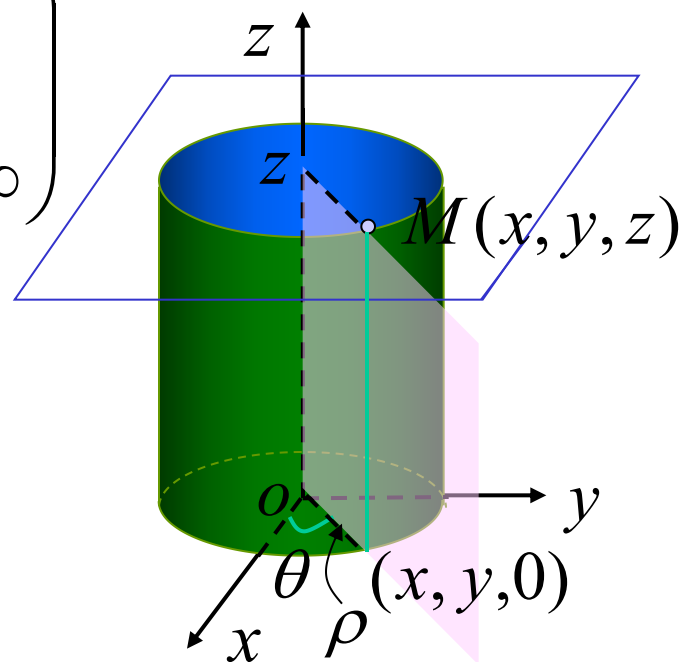
$$\begin{cases} x = \rho \cos \theta \\ y = \rho \sin \theta \\ z = z \end{cases} \quad \begin{cases} 0 \leq \rho < +\infty \\ 0 \leq \theta \leq 2\pi \\ -\infty < z < +\infty \end{cases}$$

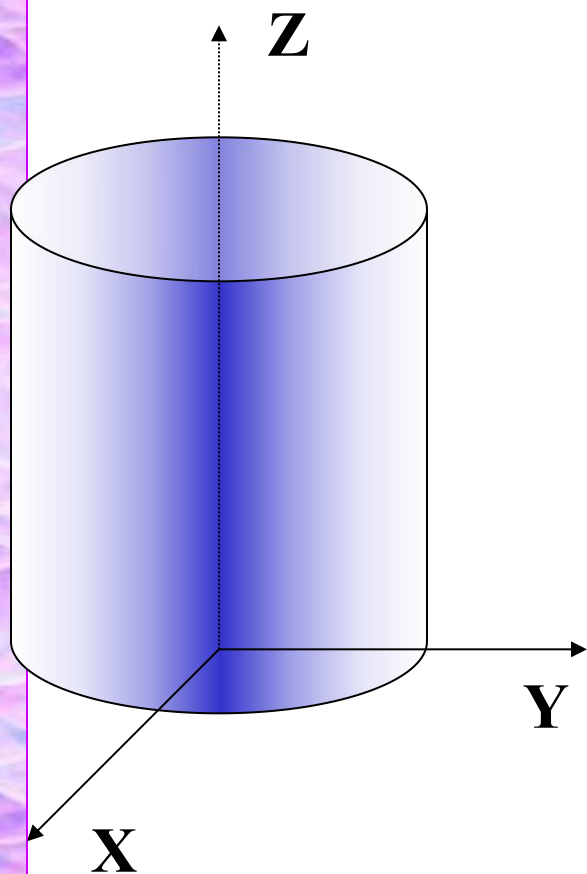
坐标面分别为

$\rho = \text{常数}$ $\longrightarrow$ 圆柱面

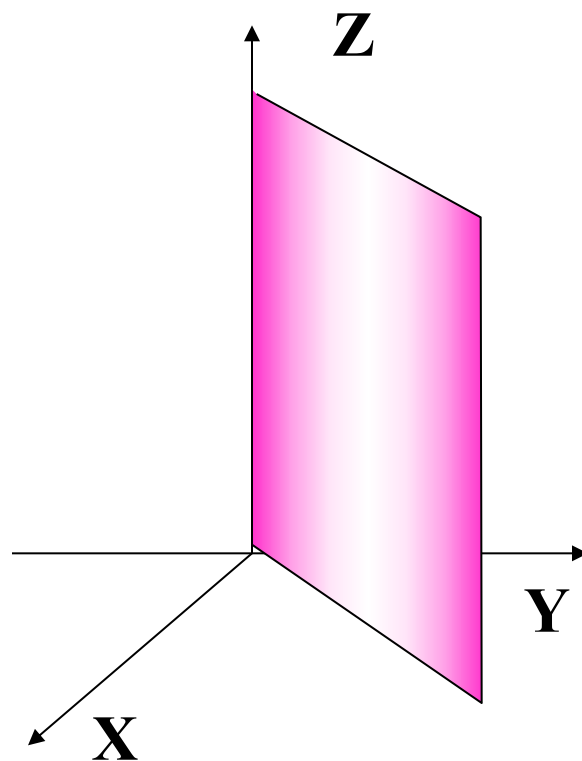
$\theta = \text{常数}$ $\longrightarrow$ 半平面

$z = \text{常数}$ $\longrightarrow$ 平面

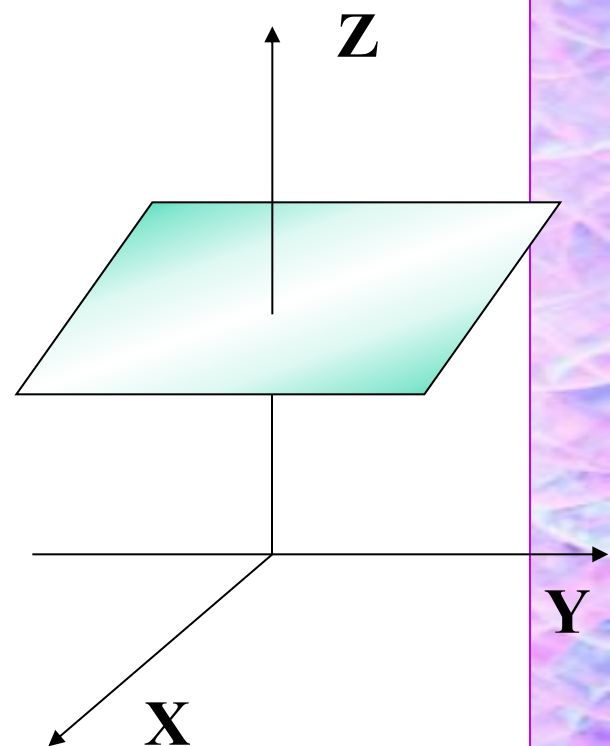




$\rho = \text{常数}$
—圆柱面



$\theta = \text{常数}$
—半平面



$z = \text{常数}$
—水平面

2. 利用柱坐标计算三重积分

如图所示, 在柱面坐标系中体积元素为

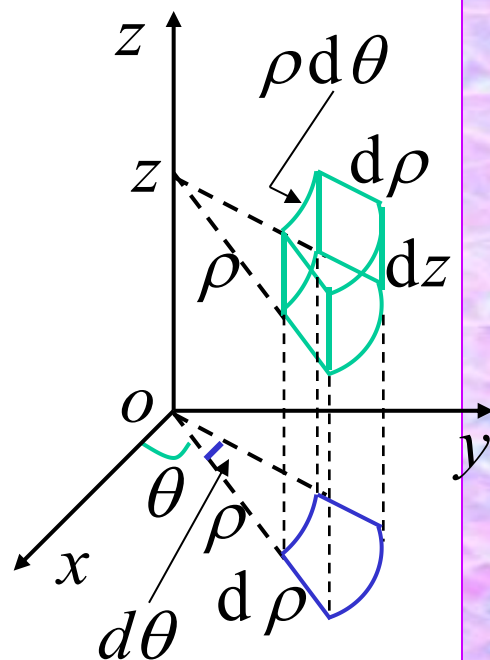
$$d v = \rho d \rho d \theta d z$$

因此

$$\iiint_{\Omega} f(x, y, z) dx dy dz$$

$$= \iiint_{\Omega} F(\rho, \theta, z) \rho d \rho d \theta dz$$

其中 $F(\rho, \theta, z) = f(\rho \cos \theta, \rho \sin \theta, z)$



适用范围:

- 1) 积分域表面用柱面坐标表示时方程简单;
- 2) 被积函数用柱面坐标表示时变量互相分离.

$$\iiint_{\Omega} f(x, y, z) dx dy dz = \iiint_{\Omega} f(\rho \cos \theta, \rho \sin \theta, z) \rho d\rho d\theta dz$$

化成三次积分为：

$$\iiint_{\Omega} = \iint_D \left[\int_{z_1(\rho, \theta)}^{z_2(\rho, \theta)} f(\rho \cos \theta, \rho \sin \theta, z) dz \right] \rho d\rho d\theta$$

$$= \int_{\alpha}^{\beta} d\theta \int_{\rho_1(\theta)}^{\rho_2(\theta)} \rho d\rho \int_{z_1(\rho, \theta)}^{z_2(\rho, \theta)} f(\rho \cos \theta, \rho \sin \theta, z) dz$$

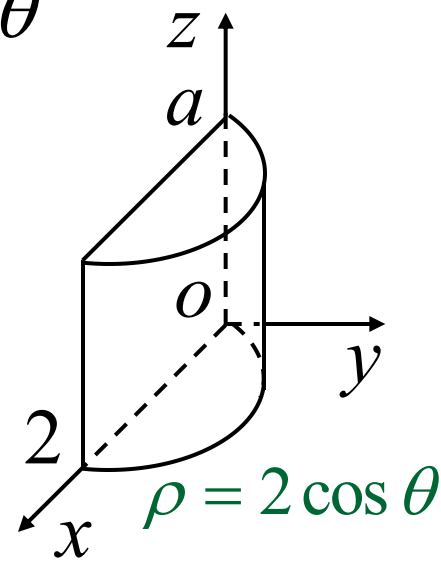
例1. 计算三重积分 $\iiint_{\Omega} z\sqrt{x^2 + y^2} \, dx dy dz$ 其中 Ω 为由柱面 $x^2 + y^2 = 2x$ 及平面 $z = 0, z = a \, (a > 0), y = 0$ 所围成半圆柱体.

解: 在柱面坐标系下 $\Omega: \begin{cases} 0 \leq \rho \leq 2 \cos \theta \\ 0 \leq \theta \leq \frac{\pi}{2} \\ 0 \leq z \leq a \end{cases}$

$$\text{原式} = \iiint_{\Omega} z \rho^2 \, d\rho d\theta dz$$

$$= \int_0^a z dz \int_0^{\pi/2} d\theta \int_0^{2 \cos \theta} \rho^2 \, d\rho$$

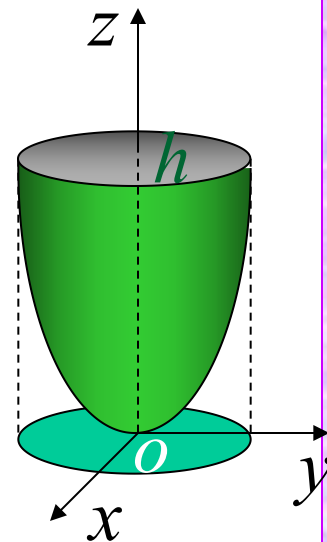
$$= \frac{4a^2}{3} \int_0^{\pi/2} \cos^3 \theta \, d\theta = \frac{8}{9} a^3$$



$$dv = \rho \, d\rho \, d\theta \, dz$$

例2. 计算三重积分 $\iiint_{\Omega} \frac{dx dy dz}{1+x^2+y^2}$, 其中 Ω 由抛物面 $x^2 + y^2 = 4z$ 与平面 $z = h$ ($h > 0$) 所围成.

解: 在柱面坐标系下 $\Omega: \begin{cases} \frac{\rho^2}{4} \leq z \leq h \\ 0 \leq \rho \leq 2\sqrt{h} \\ 0 \leq \theta \leq 2\pi \end{cases}$



$$\text{原式} = \int_0^{2\pi} d\theta \int_0^{2\sqrt{h}} \frac{\rho}{1+\rho^2} d\rho \int_{\frac{\rho^2}{4}}^h dz$$

$$= 2\pi \int_0^{2\sqrt{h}} \frac{\rho}{1+\rho^2} \left(h - \frac{\rho^2}{4}\right) d\rho$$

$$= \frac{\pi}{4} [(1+4h) \ln(1+4h) - 4h]$$

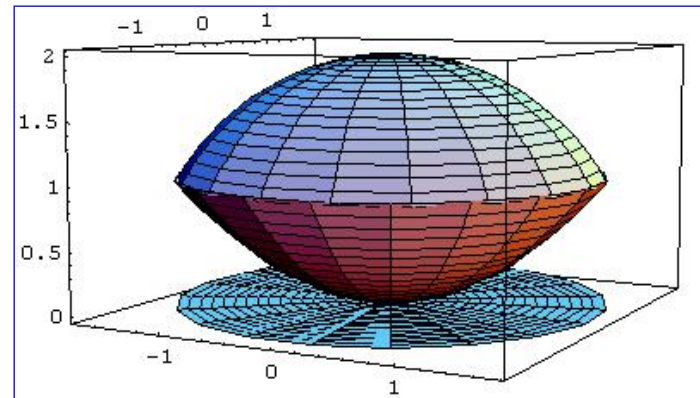
$$dv = \rho d\rho d\theta dz$$

例 3. 计算 $I = \iiint_{\Omega} z dx dy dz$, 其中 Ω 是球面

$x^2 + y^2 + z^2 = 4$ 与抛物面 $x^2 + y^2 = 3z$
所围的立体.

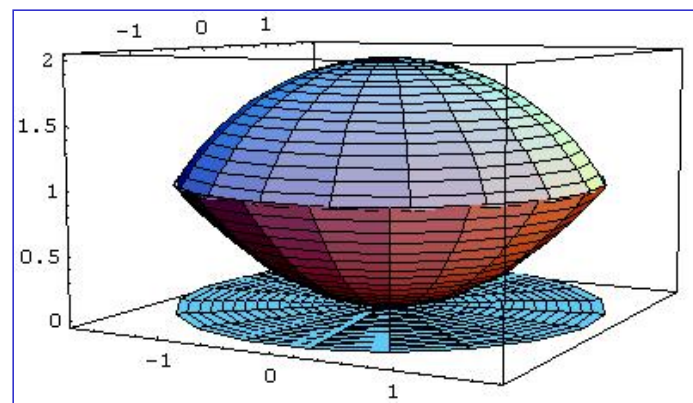
解
$$\begin{cases} x = \rho \cos \theta \\ y = \rho \sin \theta \\ z = z \end{cases}, \quad \text{知交线为}$$

$$\begin{cases} \rho^2 + z^2 = 4 \\ \rho^2 = 3z \end{cases} \Rightarrow z = 1, \quad \rho = \sqrt{3},$$



把闭区域 Ω 投影到 xoy 面上, 如图,

$$\begin{aligned}\Omega: \quad & \frac{\rho^2}{3} \leq z \leq \sqrt{4-\rho^2}, \\ & 0 \leq \rho \leq \sqrt{3}, \\ & 0 \leq \theta \leq 2\pi.\end{aligned}$$



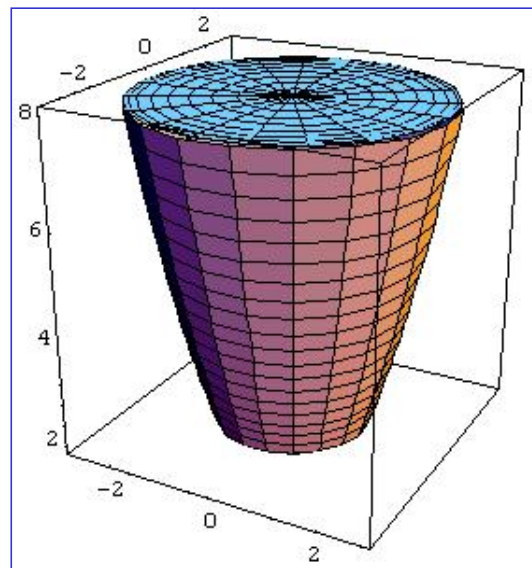
$$I = \int_0^{2\pi} d\theta \int_0^{\sqrt{3}} d\rho \int_{\frac{\rho^2}{3}}^{\sqrt{4-\rho^2}} \rho \cdot z dz = \frac{13}{4} \pi.$$

例 4 计算 $I = \iiint_{\Omega} (x^2 + y^2) dx dy dz$, 其中 Ω

是曲线 $y^2 = 2z$, $x = 0$ 绕 oz 轴旋转一周而成的曲面与两平面 $z = 2, z = 8$ 所围的立体.

解 由 $\begin{cases} y^2 = 2z \\ x = 0 \end{cases}$ 绕 oz 轴旋转得,

旋转面方程为 $x^2 + y^2 = 2z$,
所围成的立体如图,



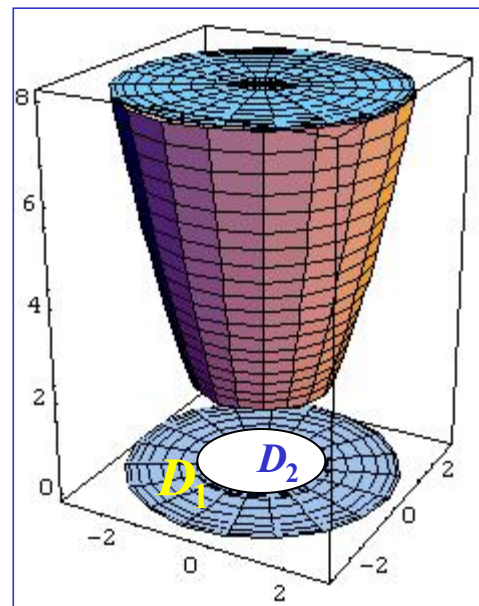
所围成立体的投影区域如图，

$$D_1 : x^2 + y^2 = 16,$$

$$\Omega_1 : \begin{cases} 0 \leq \theta \leq 2\pi \\ 0 \leq \rho \leq 4 \\ \frac{\rho^2}{2} \leq z \leq 8 \end{cases},$$

$$D_2 : x^2 + y^2 = 4,$$

$$\Omega_2 : \begin{cases} 0 \leq \theta \leq 2\pi \\ 0 \leq \rho \leq 2 \\ \frac{\rho^2}{2} \leq z \leq 2 \end{cases}.$$



$$\begin{aligned}\therefore I &= I_1 - I_2 \\ &= \iiint_{\Omega_1} (x^2 + y^2) dx dy dz - \iiint_{\Omega_2} (x^2 + y^2) dx dy dz,\end{aligned}$$

$$I_1 = \iint_{D_1} \rho d\rho d\theta \int_{\frac{\rho^2}{2}}^8 f dz = \int_0^{2\pi} d\theta \int_0^4 d\rho \int_{\frac{\rho^2}{2}}^8 \rho \cdot \rho^2 dz = \frac{4^5}{3} \pi,$$

$$I_2 = \iint_{D_2} \rho d\rho d\theta \int_{\frac{\rho^2}{2}}^2 f dz = \int_0^{2\pi} d\theta \int_0^2 d\rho \int_{\frac{\rho^2}{2}}^2 \rho \cdot \rho^2 dz = \frac{2^5}{6} \pi,$$

$$\text{原式 } I = \frac{4^5}{3} \pi - \frac{2^5}{6} \pi = 336\pi.$$

二、利用球坐标计算三重积分

1. 球面坐标

设 $M(x, y, z) \in \mathbb{R}^3$, 其柱坐标为 (ρ, θ, z) , 令 $|\overrightarrow{OM}| = r$, $\angle ZOM = \varphi$, 则 (r, φ, θ) 就称为点 M 的球坐标.

直角坐标与球面坐标的关系

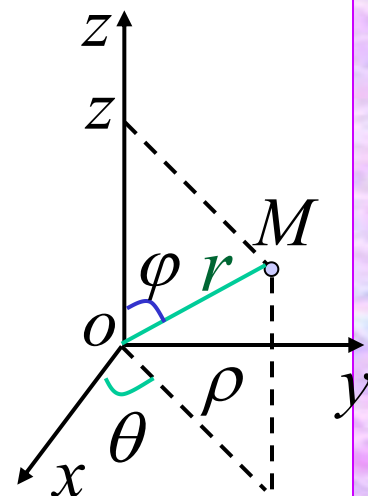
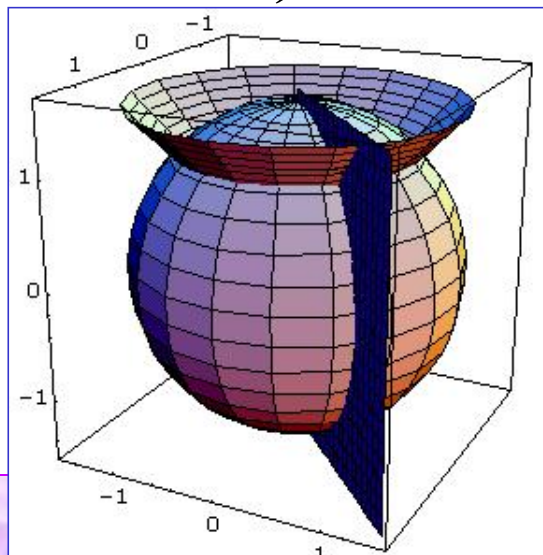
$$\begin{cases} x = r \sin \varphi \cos \theta \\ y = r \sin \varphi \sin \theta \\ z = r \cos \varphi \end{cases} \quad \begin{pmatrix} 0 \leq r < +\infty \\ 0 \leq \theta < 2\pi \\ 0 \leq \varphi \leq \pi \end{pmatrix}$$

坐标面分别为

$r = \text{常数}$ $\longrightarrow$ 球面

$\theta = \text{常数}$ $\longrightarrow$ 半平面

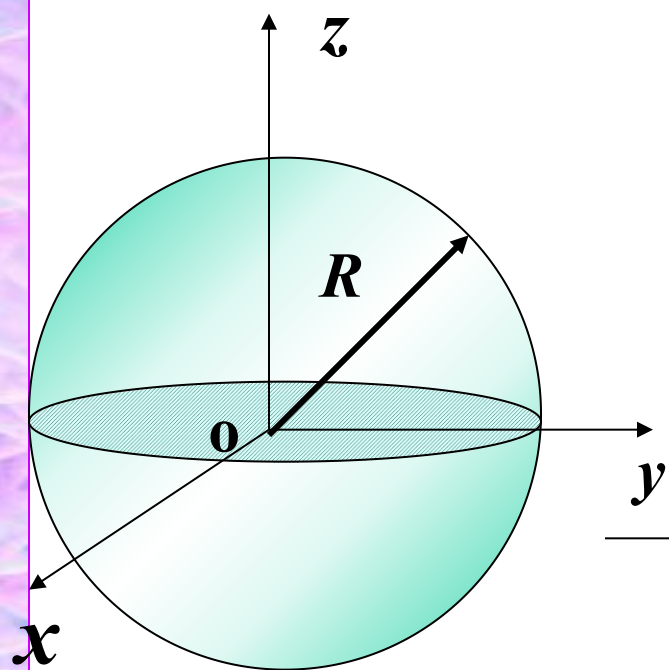
$\varphi = \text{常数}$ $\longrightarrow$ 圆锥面



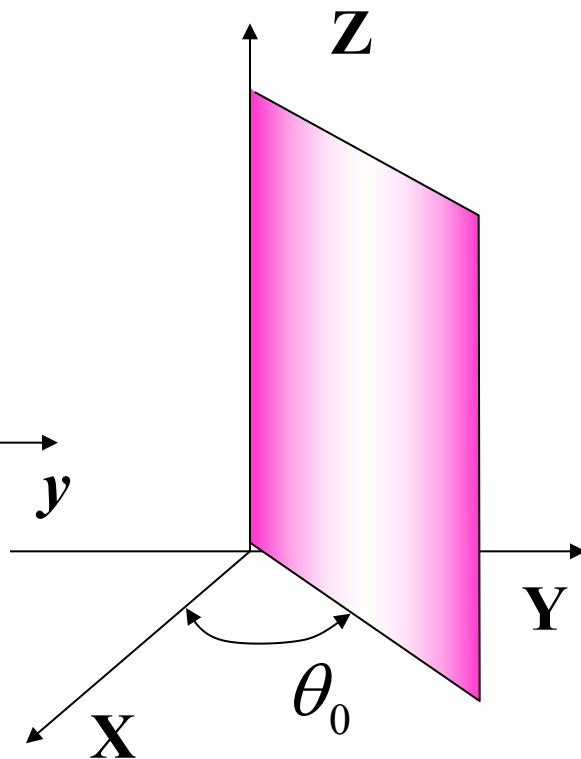
$$\rho = r \sin \varphi$$

$$z = r \cos \varphi$$

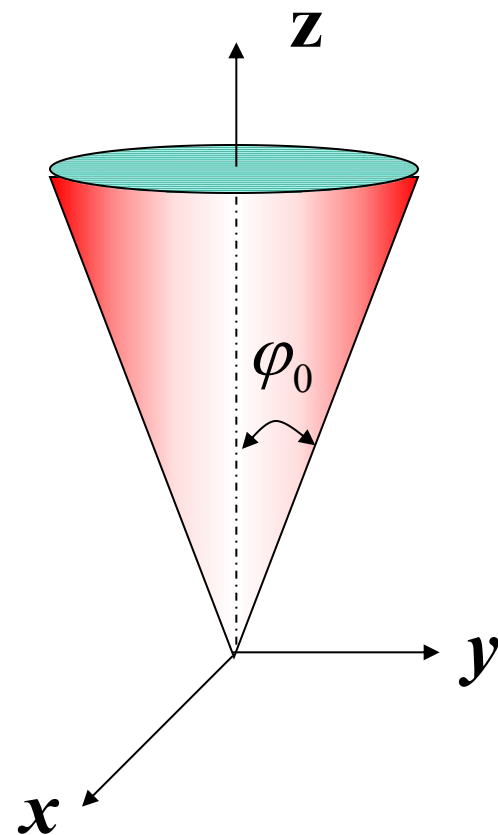
三组坐标面是：



$r = \text{常数}$
— (以 o 为球心 R 为
半径的) 球面



$\theta = \text{常数}$
— 半平面



$\varphi = \text{常数}$ — (顶点在
 o , z 轴是对称轴, 半顶
角为 φ) 圆锥面

2. 利用球面坐标计算三重积分

设 $\iiint_{\Omega} f(x, y, z) dx dy dz \exists$,

用球面坐标的三组坐标面把 Ω 分成许多小区域 

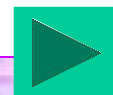
$$\Delta v \approx AB \cdot AC \cdot AD$$

$$AB = r d\varphi \quad AC = |AE| d\theta = r \sin \varphi d\theta \quad AD = dr$$

$$\Rightarrow dv = r^2 \sin \varphi dr d\varphi d\theta$$

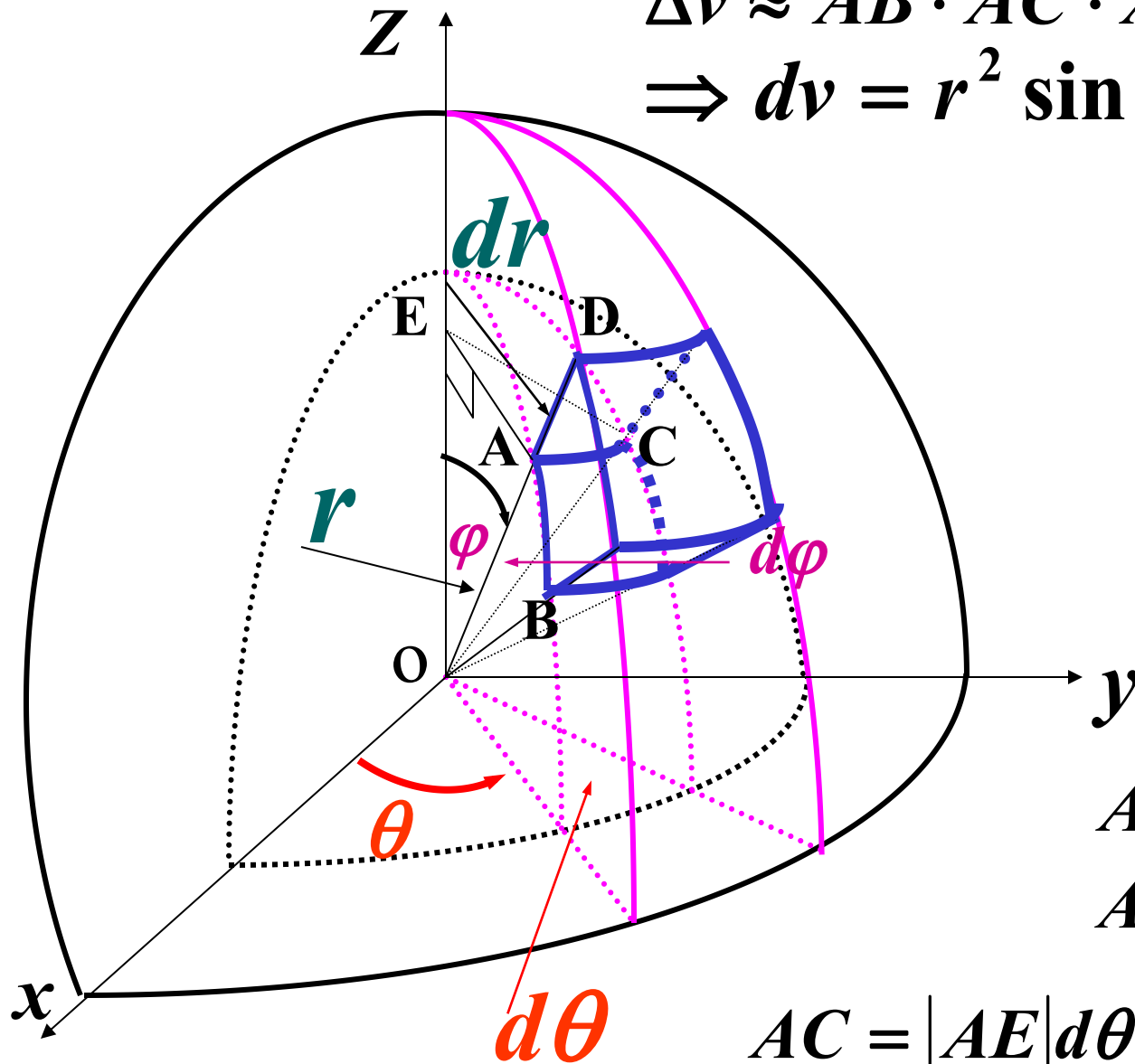
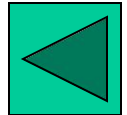
$$\therefore \iiint_{\Omega} f(x, y, z) dx dy dz$$

$$= \iiint_{\Omega} f(r \sin \varphi \cos \theta, r \sin \varphi \sin \theta, r \cos \varphi) \cdot r^2 \sin \varphi dr d\varphi d\theta$$



$$\Delta v \approx AB \cdot AC \cdot AD$$

$$\Rightarrow dv = r^2 \sin \varphi dr d\varphi d\theta$$



$$AB = r d\varphi$$

$$AD = dr$$

$$AC = |AE| d\theta = r \sin \varphi d\theta$$

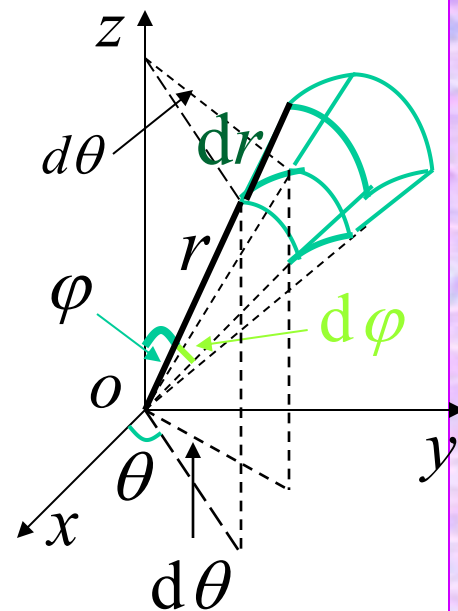
如图所示, 在球面坐标系中体积元素为

$$d v = r^2 \sin \varphi d r d \varphi d \theta$$

因此有

$$\iiint_{\Omega} f(x, y, z) d x d y d z$$

$$= \iiint_{\Omega} F(r, \theta, \varphi) r^2 \sin \varphi d r d \varphi d \theta$$



其中 $F(r, \theta, \varphi) = f(r \sin \varphi \cos \theta, r \sin \varphi \sin \theta, r \cos \varphi)$

适用范围:

- 1) 积分域表面用球面坐标表示时方程简单;
- 2) 被积函数用球面坐标表示时变量互相分离.

$$\iiint_{\Omega} f(x, y, z) dx dy dz$$

$$= \iiint_{\Omega} F(r, \theta, \varphi) r^2 \sin \varphi dr d\varphi d\theta$$

其中 $F(r, \theta, \varphi) = f(r \sin \varphi \cos \theta, r \sin \varphi \sin \theta, r \cos \varphi)$

当 Ω 为球面 $r = a$

$$I = \int_0^{2\pi} d\theta \int_0^{\pi} d\varphi \int_0^a F(r, \varphi, \theta) r^2 \sin \varphi dr$$

当 $F(r, \varphi, \theta) = 1$

$$I = V_{\text{球}} = \int_0^{2\pi} d\theta \int_0^{\pi} d\varphi \int_0^a r^2 \sin \varphi dr = \frac{4}{3} \pi a^3$$

例 5 求曲面 $x^2 + y^2 + z^2 = 2a^2$ 与 $z = \sqrt{x^2 + y^2}$ 所围成的立体体积.

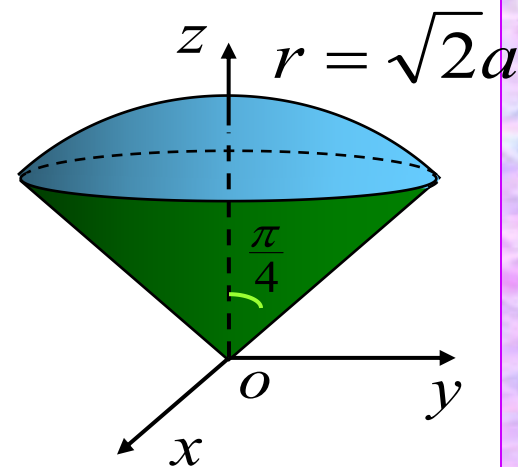
解 Ω 由锥面和球面围成, 采用球面坐标,

$$\text{由 } x^2 + y^2 + z^2 = 2a^2$$

$$\Rightarrow r = \sqrt{2}a,$$

$$z = \sqrt{x^2 + y^2} \Rightarrow \varphi = \frac{\pi}{4},$$

$$\Omega: \quad 0 \leq r \leq \sqrt{2}a, \quad 0 \leq \varphi \leq \frac{\pi}{4}, \quad 0 \leq \theta \leq 2\pi,$$

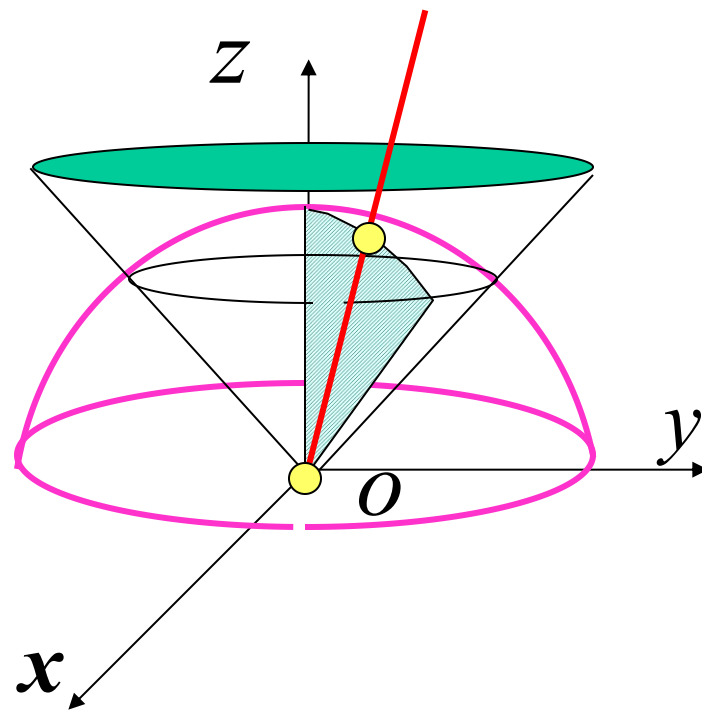


由三重积分的性质知 $V = \iiint_{\Omega} dx dy dz,$

$$V = \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{4}} d\varphi \int_0^{\sqrt{2}a} r^2 \sin \varphi dr$$

$$= 2\pi \int_0^{\frac{\pi}{4}} \sin \varphi \cdot \frac{(\sqrt{2}a)^3}{3} d\varphi$$

$$= \frac{4}{3} \pi (\sqrt{2} - 1) a^3.$$



例6. 计算三重积分 $\iiint_{\Omega} \sqrt{x^2 + y^2 + z^2} dv$, 其中

(1) Ω 是由球面 $x^2 + y^2 + z^2 = z$ 所围成的闭区域;

解 在球面坐标系中, 球面 $x^2 + y^2 + z^2 = z$ 的方程为 $r^2 = r \cos \varphi$, 即 $r = \cos \varphi$. Ω 可表示为 $0 \leq r \leq \cos \varphi, 0 \leq \varphi \leq \frac{\pi}{2}, 0 \leq \theta \leq 2\pi$ (图 10 - 46).

$$\begin{aligned}\iiint_{\Omega} \sqrt{x^2 + y^2 + z^2} dv &= \iiint_{\Omega} r \cdot r^2 \sin \varphi dr d\varphi d\theta \\&= \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{2}} \sin \varphi d\varphi \int_0^{\cos \varphi} r^3 dr \\&= 2\pi \int_0^{\frac{\pi}{2}} \sin \varphi \cdot \frac{\cos^4 \varphi}{4} d\varphi \\&= -\frac{\pi}{2} \left[\frac{\cos^5 \varphi}{5} \right]_0^{\frac{\pi}{2}} = \frac{\pi}{10}.\end{aligned}$$

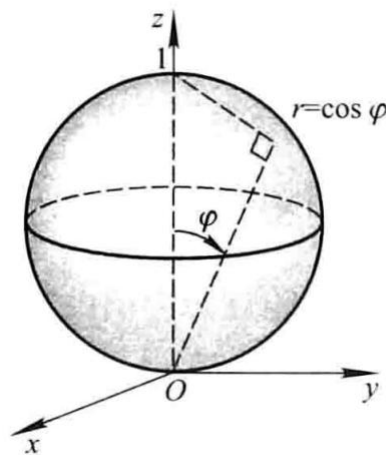


图 10 - 46

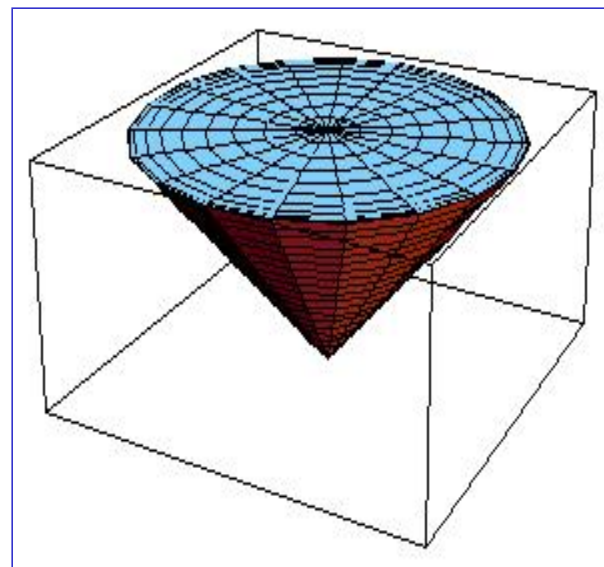
例 7 计算 $I = \iiint_{\Omega} (x^2 + y^2) dx dy dz$, 其中 Ω 是锥面 $x^2 + y^2 = z^2$, 与平面 $z = a$ ($a > 0$) 所围的立体.

解 1 采用球面坐标

$$\because z = a \Rightarrow r = \frac{a}{\cos \varphi},$$

$$x^2 + y^2 = z^2 \Rightarrow \varphi = \frac{\pi}{4},$$

$$\therefore \Omega: 0 \leq r \leq \frac{a}{\cos \varphi}, \quad 0 \leq \varphi \leq \frac{\pi}{4}, \quad 0 \leq \theta \leq 2\pi,$$



$$I = \iiint_{\Omega} (x^2 + y^2) dx dy dz$$

$$= \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{4}} d\varphi \int_0^{\frac{a}{\cos \varphi}} r^4 \sin^3 \varphi dr$$

$$= 2\pi \int_0^{\frac{\pi}{4}} \sin^3 \varphi \cdot \frac{1}{5} \left(\frac{a^5}{\cos^5 \varphi} - 0 \right) d\varphi$$

$$= \frac{\pi}{10} a^5.$$

解 2 采用柱面坐标

$$\because x^2 + y^2 = z^2 \Rightarrow z = \rho, \quad D: x^2 + y^2 \leq a^2,$$

$$\Omega: \rho \leq z \leq a, \quad 0 \leq \rho \leq a, \quad 0 \leq \theta \leq 2\pi,$$

$$\begin{aligned} I &= \iiint_{\Omega} (x^2 + y^2) dx dy dz = \int_0^{2\pi} d\theta \int_0^a d\rho \int_{\rho}^a \rho^3 dz \\ &= 2\pi \int_0^a \rho^3 (a - \rho) d\rho = 2\pi \left[a \cdot \frac{a^4}{4} - \frac{a^5}{5} \right] = \frac{\pi}{10} a^5. \end{aligned}$$

复习 直线 L 绕另一条与 L 相交的直线旋转一周，所得旋转曲面叫**圆锥面**．两直线的交点叫圆锥面的**顶点**，两直线的夹角 $\alpha \left(0 < \alpha < \frac{\pi}{2} \right)$ 叫圆锥面的**半顶角**．试建立顶点在坐标原点，旋转轴为 z 轴，半顶角为 α 的圆锥面方程．

解 $yo z$ 面上直线方程为

$$z = y \cot \alpha$$

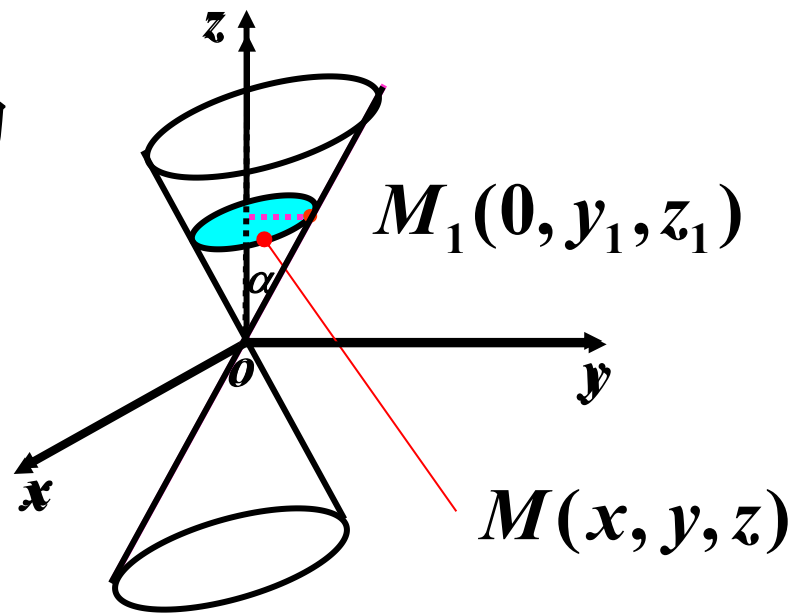
圆锥面方程

$$z = \pm \sqrt{x^2 + y^2} \cot \alpha$$

令 $a = \cot \alpha$

两边平方

$$z^2 = \cot^2 \alpha (x^2 + y^2)$$



例8 球心在 $(0,0,a)$,半径为 a 的球与半顶角为 α 的内接锥面所围成的立体体积

解

直角坐标系方程

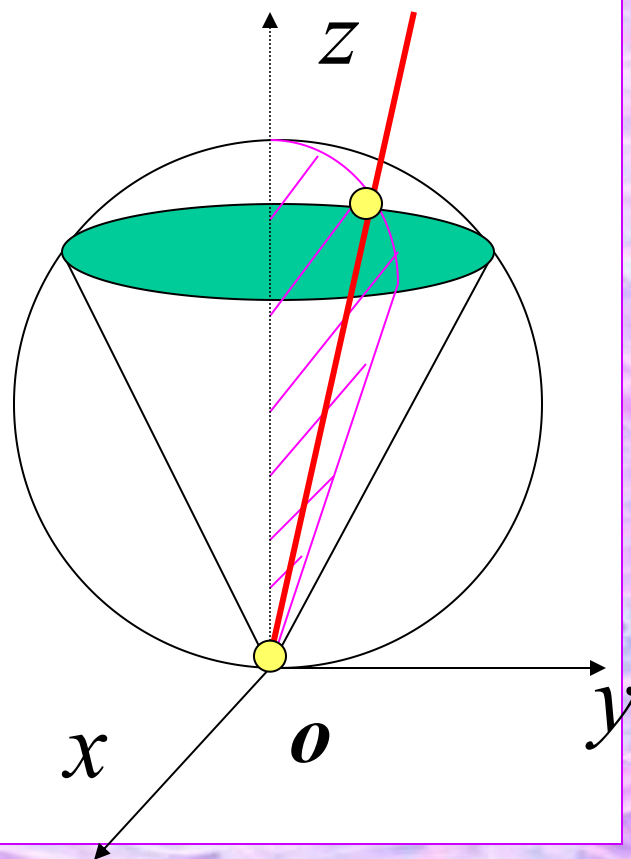
$$\text{球面: } x^2 + y^2 + (z - a)^2 = a^2,$$

$$\text{锥面: } z^2 = \cot^2 \alpha (x^2 + y^2)$$

球坐标系方程:

$$\text{球面: } r = 2a \cos \phi$$

$$\text{锥面: } \phi = \alpha$$



例8 球心在 $(0,0,a)$,半径为 a 的球与半顶角为 α 的内接锥面所围成的立体体积

解 球面： $r = 2a \cos \varphi$ 锥面： $\varphi = \alpha$

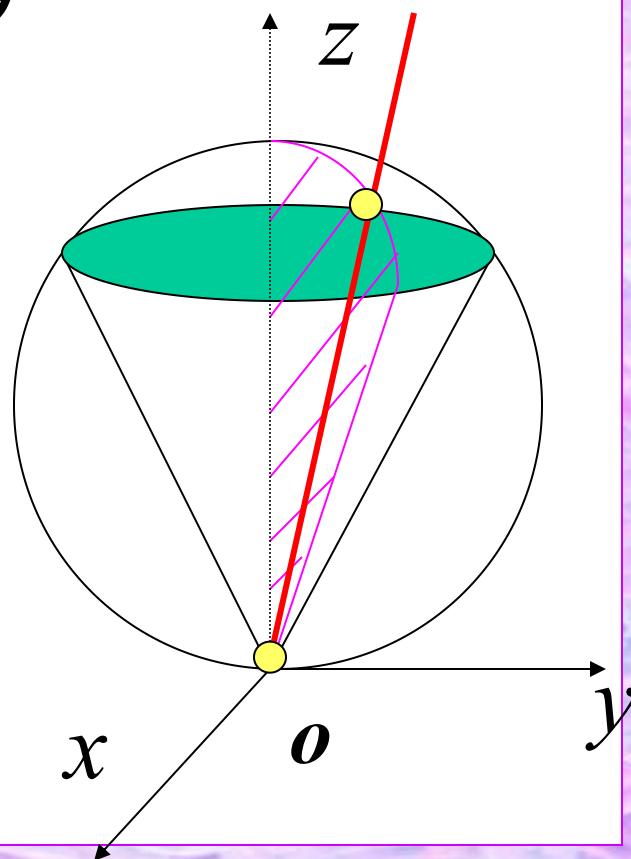
$$V = \iiint_{\Omega} dx dy dz = \iiint_{\Omega} r^2 \sin \varphi dr d\varphi d\theta$$

$$= \int_0^{2\pi} d\theta \int_0^{\alpha} d\varphi \int_0^{2a \cos \varphi} r^2 \sin \varphi dr$$

$$= \int_0^{2\pi} d\theta \int_0^{\alpha} \frac{2^3}{3} a^3 \cos^3 \varphi \sin \varphi d\varphi$$

$$= 2\pi \frac{2^3}{3} a^3 \left[-\frac{1}{4} \cos^4 \varphi \right]_0^{\alpha}$$

$$= \frac{4\pi}{3} a^3 [1 - \cos^4 \alpha]$$



例9 计算 $I = \iiint_{\Omega} z dx dy dz$, 其中 Ω 是

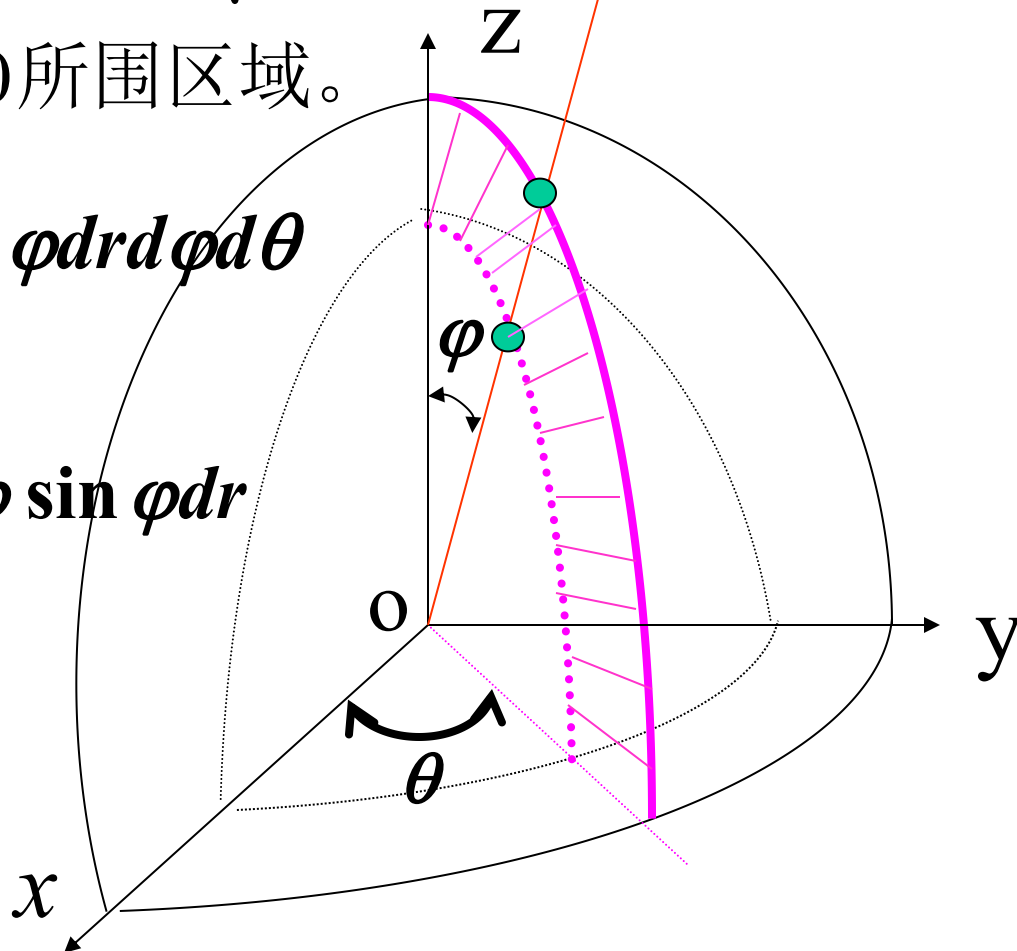
$$z = \sqrt{A^2 - x^2 - y^2}, z = \sqrt{a^2 - x^2 - y^2}$$

$(A > a > 0), z = 0$ 所围区域。

解 $I = \iiint_{\Omega} r \cos \varphi r^2 \sin \varphi dr d\varphi d\theta$

$$= \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{2}} d\varphi \int_a^A r^3 \cos \varphi \sin \varphi dr$$

$$= \frac{\pi}{4} (A^4 - a^4)$$



三重积分的对称性和轮换对称性

复习： 定积分偶倍奇零性质

当 $f(x)$ 在 $[-a, a]$ 上连续, 且有

① $f(x)$ 为偶函数, 则

$$\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx;$$

② $f(x)$ 为奇函数, 则 $\int_{-a}^a f(x) dx = 0$.

思考题：

将定积分偶倍奇零性质推广到三重积分的情形

补充：利用对称性化简三重积分计算

使用对称性时应注意：

- 1、积分区域关于坐标面的对称性；
- 2、被积函数在积分区域上的关于三个坐标轴的奇偶性。

一般地，当积分区域 Ω 关于 xoy 平面对称，且被积函数 $f(x, y, z)$ 是关于 z 的奇函数，则三重积分为零，若被积函数 $f(x, y, z)$ 是关于 z 的偶函数，则三重积分为 Ω 在 xoy 平面上方的半个闭区域的三重积分的两倍。

例 10 利用对称性简化计算

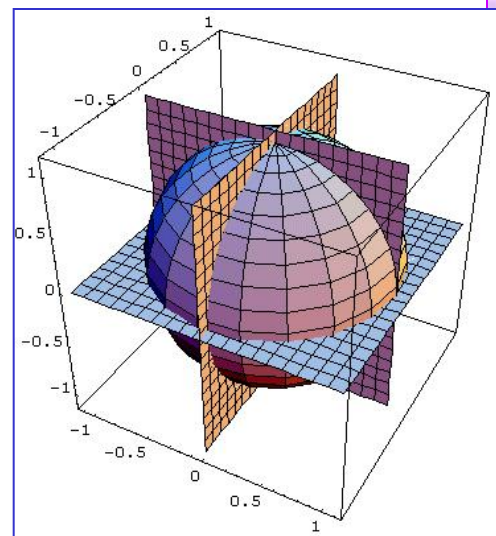
$$\iiint_{\Omega} \frac{z \ln(x^2 + y^2 + z^2 + 1)}{x^2 + y^2 + z^2 + 1} dx dy dz$$

其中积分区域 $\Omega = \{(x, y, z) \mid x^2 + y^2 + z^2 \leq 1\}$.

解 积分域关于三个坐标面都对称,

被积函数是 z 的奇函数,

$$\iiint_{\Omega} \frac{z \ln(x^2 + y^2 + z^2 + 1)}{x^2 + y^2 + z^2 + 1} dx dy dz = 0.$$



例11 求 $\iiint_{\Omega} e^{|z|} dv$, 其中 $\Omega: x^2 + y^2 + z^2 \leq 1$

解法一 $\iiint_{\Omega} e^{|z|} dv = \int_{-1}^1 e^{|z|} dz \iint_{D_z} dx dy \quad D_z: x^2 + y^2 \leq 1 - z^2$

$$= \pi \int_{-1}^1 e^{|z|} (1 - z^2) dz = 2\pi \int_0^1 e^z (1 - z^2) dz = 2\pi$$

解法二 $\Omega_1 : x^2 + y^2 + z^2 \leq 1$ 及 $z \geq 0$

$$D: x^2 + y^2 \leq 1$$

$$\iiint_{\Omega} e^{|z|} dv = 2 \iiint_{\Omega_1} e^z dv$$

$$= 2 \iint_D dx dy \int_0^{\sqrt{1-x^2-y^2}} e^z dz = 2 \iint_D (e^{\sqrt{1-x^2-y^2}} - 1) dx dy$$

$$= 2 \int_0^{2\pi} d\theta \int_0^1 (e^{\sqrt{1-r^2}} - 1) r dr$$

$$= 4\pi \int_0^1 e^{\sqrt{1-r^2}} r dr - 2\pi$$

$$\begin{aligned} & \text{令 } \sqrt{1-r^2} = t \\ & = 4\pi \int_0^1 e^t t dt - 2\pi = 2\pi \end{aligned}$$

解法三 $\iiint_{\Omega} e^{|z|} dv = 2 \iiint_{\Omega_1} e^z dv$

$$\Omega_1 : x^2 + y^2 + z^2 \leq 1 \text{ 及 } z \geq 0$$

$$\text{原式} = 2 \iiint_{\Omega_1} e^{r \cos \varphi} r^2 \sin \varphi dr d\varphi d\theta$$

$$= 2 \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{2}} d\varphi \int_0^1 e^{r \cos \varphi} r^2 \sin \varphi dr$$

$$= 4\pi \int_0^{\frac{\pi}{2}} \sin \varphi d\varphi \int_0^1 e^{r \cos \varphi} r^2 dr$$

$$= 4\pi \int_0^{\frac{\pi}{2}} \sin \varphi e^{\cos \varphi} \left[\frac{1}{\cos \varphi} - \frac{2}{\cos^2 \varphi} - \frac{2}{\cos^3 \varphi} \right] d\varphi$$

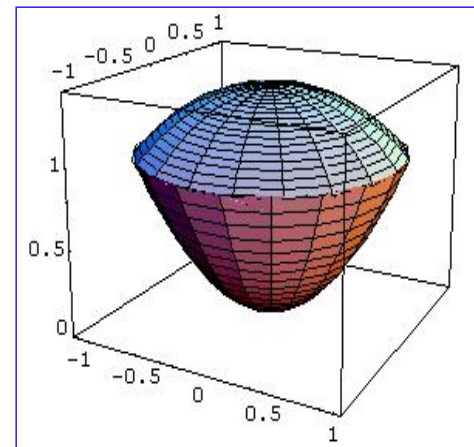
$$= \cdots = 2\pi$$

例 12 计算 $\iiint_{\Omega} (x+y+z)^2 dx dy dz$ 其中 Ω 是由抛物面 $z = x^2 + y^2$ 和球面 $x^2 + y^2 + z^2 = 2$ 所围成的空间闭区域.

解 $\because (x+y+z)^2$
 $= x^2 + y^2 + z^2 + 2(\underline{xy + yz + zx})$

其中 $xy + yz$ 是关于 y 的奇函数,

且 Ω 关于 zox 面对称, $\therefore \iiint_{\Omega} (xy + yz) dv = 0,$



同理 $\because zx$ 是关于 x 的奇函数,

且 Ω 关于 $yo z$ 面对称, $\therefore \iiint_{\Omega} xz d\nu = 0,$

由对称性知 $\iiint_{\Omega} x^2 d\nu = \iiint_{\Omega} y^2 d\nu,$

$$\begin{aligned} \text{则 } I &= \iiint_{\Omega} (x + y + z)^2 dx dy dz \\ &= \iiint_{\Omega} (2x^2 + z^2) dx dy dz, \end{aligned}$$

在柱面坐标下：

$$0 \leq \theta \leq 2\pi, \quad 0 \leq \rho \leq 1, \quad \rho^2 \leq z \leq \sqrt{2 - \rho^2},$$

投影区域 D_{xy} : $x^2 + y^2 \leq 1$,

$$\begin{aligned} I &= \int_0^{2\pi} d\theta \int_0^1 d\rho \int_{\rho^2}^{\sqrt{2-\rho^2}} \rho(2\rho^2 \cos^2 \theta + z^2) dz \\ &= \frac{\pi}{60} (90\sqrt{2} - 89). \end{aligned}$$

三重积分的轮换对称性

例 13. 设 $\Omega = \{(x, y, z) \mid x^2 + y^2 + z^2 \leq a^2\}$,
计算 $\iiint_{\Omega} (lx^2 + my^2 + nz^2) dv$ (其中 l, m, n 为常数).

解 由轮换对称性

$$\begin{aligned} \iiint_{\Omega} (lx^2 + my^2 + nz^2) dv &= (l + m + n) \iiint_{\Omega} z^2 dv \\ &= (l + m + n) \int_{-a}^a z^2 dz \iint_{D_z} dx dy \\ &= (l + m + n) \int_{-a}^a \pi z^2 (a^2 - z^2) dz \\ &= \frac{4}{15} \pi a^5 (l + m + n) \end{aligned}$$

$D_z : x^2 + y^2 \leq a^2 - z^2$

例 14. 求 $F'(t)$, 设

$$(2) F(t) = \iiint_{x^2+y^2+z^2 \leq t^2} f(x^2 + y^2 + z^2) dV, \quad \text{其中 } f(u) \text{ 为可微函数};$$

解

$$\begin{aligned} F(t) &= \iiint_{x^2+y^2+z^2 \leq t^2} f(x^2 + y^2 + z^2) dV \\ &= \int_0^{2\pi} d\theta \int_0^\pi \sin \varphi d\varphi \int_0^t f(r^2) r^2 dr \\ &= 4\pi \int_0^t f(r^2) r^2 dr \end{aligned}$$

$$F'(t) = 4\pi t^2 f(t^2)$$

三重积分的换元法

空间上同一个点，直角坐标与球坐标之

间的关系为

$$\begin{cases} x = r \sin \varphi \cos \theta \\ y = r \sin \varphi \sin \theta, \\ z = r \cos \varphi. \end{cases}$$

上式可看成是从球坐标到直角坐标的一种变换，
且这种变换是一对一的。

定理 设 $f(x, y, z)$ 在空间区域 V 连续, 作变换

$$T: \quad x = x(u, v, w), \quad y = y(u, v, w), \quad z = z(u, v, w)$$

满足

(1) $x(u, v, w), y(u, v, w), z(u, v, w)$ 具有一阶连续偏导数;

$$(2) \text{ 雅可比式 } J(u, v, w) = \frac{\partial(x, y, z)}{\partial(u, v, w)} = \begin{vmatrix} x_u & x_v & x_w \\ y_u & y_v & y_w \\ z_u & z_v & z_w \end{vmatrix} \neq 0;$$

则有

$$\iiint_D f(x, y, z) dx dy dz = \iiint_D f[x(u, v, w), y(u, v, w), z(u, v, w)] |J(u, v, w)| du dv dw.$$

作球坐标变换

$$\begin{cases} x = r \sin \varphi \cos \theta, \\ y = r \sin \varphi \sin \theta, \\ z = r \cos \varphi. \end{cases}$$

$$J = \frac{\partial(x, y, z)}{\partial(r, \varphi, \theta)} = \begin{vmatrix} x_r & x_\varphi & x_\theta \\ y_r & y_\varphi & y_\theta \\ z_r & z_\varphi & z_\theta \end{vmatrix}$$

$$= \begin{vmatrix} \sin \varphi \cos \theta & r \cos \varphi \cos \theta & -r \sin \varphi \sin \theta \\ \sin \varphi \sin \theta & r \cos \varphi \sin \theta & r \sin \varphi \cos \theta \\ \cos \varphi & -r \sin \varphi & 0 \end{vmatrix}$$

$$= r^2 \sin \varphi \cos^2 \varphi + r^2 \sin^3 \varphi = r^2 \sin \varphi$$

例15. 求 $I = \iiint_{\Omega} \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \right) dx dy dz$, $\Omega: \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \leq 1$.

解: 作广义球坐标变换
$$\begin{cases} x = ar \sin \theta \cos \varphi, \\ y = br \sin \theta \sin \varphi, \\ z = cr \cos \theta. \end{cases}$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = r^2 \Rightarrow r \leq 1$$

$$I = \iiint_{\Omega} \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \right) dx dy dz = \iiint_{\Omega'} r^2 |J| dr d\theta d\varphi$$

$$J = abcr^2 \sin \theta$$

$$\therefore I = abc \int_0^{2\pi} d\varphi \int_0^{\pi} \sin \theta d\theta \int_0^1 r^4 dr = \frac{4}{5} \pi abc.$$

解法二(用截面法)

$$\begin{aligned}\iiint_{\Omega} \left(\frac{x^2}{a^2}\right) dv &= \int_{-a}^a \frac{x^2}{a^2} dx \iint_{D_x} dy dz \\&= \int_{-a}^a \frac{x^2}{a^2} \pi bc \left(1 - \frac{x^2}{a^2}\right) dx \\&= 2 \int_0^a \frac{x^2}{a^2} \pi bc \left(1 - \frac{x^2}{a^2}\right) dx \\&= 2\pi bc \left[\frac{x^3}{3a^2} - \frac{x^5}{5a^4} \right]_0^a = \frac{4}{15} \pi abc\end{aligned}$$

$$\text{同理, } \iiint_{\Omega} \left(\frac{y^2}{b^2}\right) dv = \iiint_{\Omega} \left(\frac{z^2}{c^2}\right) dv = \frac{4}{15} \pi abc$$

$$D_x : \frac{y^2}{b^2} + \frac{z^2}{c^2} \leq 1 - \frac{x^2}{a^2},$$

$$S_x = \pi bc \left(1 - \frac{x^2}{a^2}\right)$$

$$\begin{aligned}\therefore \iiint_{\Omega} \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2}\right) dv \\&= \frac{4}{5} \pi abc\end{aligned}$$

内容小结

坐标系	体积元素	适用情况
直角坐标系	$dx dy dz$	积分区域多由坐标面围成； 被积函数形式简洁，或 变量可分离.
柱面坐标系	$\rho d\rho d\varphi dz$	
球面坐标系	$r^2 \sin\theta dr d\varphi d\theta$	

* 说明：

三重积分也有类似二重积分的换元积分公式：

$$\iiint_{\Omega} f(x, y, z) dx dy dz = \iiint_{\Omega}^* F(u, v, w) |J| du dv dw$$

对应雅可比行列式为 $J = \frac{\partial(x, y, z)}{\partial(u, v, w)}$

小结

三重积分换元法 $\left\{ \begin{array}{l} \text{柱面坐标} \\ \text{球面坐标} \end{array} \right.$

(1) 柱面坐标的体积元素

$$dxdydz = r dr d\theta dz$$

(2) 球面坐标的体积元素

$$dxdydz = r^2 \sin \varphi dr d\theta d\varphi$$

(3) 对称性简化运算